

Backside Power Delivery in High Density and High Performance Context: IR-drop and Block-level Power-Performance-Area Benefits

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Abstract – We evaluate block-level power-performance-area (PPA) tradeoffs of two backside power (BSPDN) options: Through Silicon Via in the Middle of Line (TSVM) and Backside Contact (BSC) in an A10 (10Å) nanosheet technology node. The benchmarking was conducted for high-performance designs in both high-performance and high-density technology scenarios taking the traditional frontside power option as the baseline. HP technology clearly benefits most since its dense PDN consumes a lot of metal routing resources on the frontside to keep an acceptable IR-drop. Introducing BSPDN frees up those metals for more efficient signal routing, resulting in 19% smaller area (8% for HD). Continued standard cell scaling by means of BSC translates into a further -25% core area, with only the densest 4.5T HD results showing signs of congestion at high frequencies.

Keywords: Backside power, TSVM, backside contact, nanosheet

Introduction

The growing demands from data centers, mobile cores, and artificial intelligence continuously drive CMOS technology development towards advanced nodes for PPA improvements. Backside power delivery has been proposed as a structural scaling booster for enhanced PPA efficiency [1-4]. Various backside power connections were proposed with device-level PPA analyzed in HD and HP technologies [3]. A block-level PPA study has been performed with the buried power rail (BPR) option in HP [4]. However, the potential of TSVM and BSC at block level in HP and HD vs. frontside power delivery remains unknown. A comprehensive PPA study with physical design experiments is necessary to reveal the PPA tradeoffs. We therefore evaluate these backside power solutions in terms of PPA, taking the traditional frontside power delivery (FS M0) as the baseline (Fig. 1(a)).

Technology and Benchmarking Methodology

Backside power delivery network directly delivers power to active devices from the wafer's backside. For TSVM, a tall via (Fig. 1(b)) in the Middle-Of-Line is used to connect the side of CT (contact to epi) to the first backside metal layer [3]. For BSC, a via directly connects to the bottom of the active device from the backside [3]. We leverage the most efficient implementation of BSC, where the via becomes a wrap-around contact to the device (Fig. 1(c)). As such, the contact resistance is reduced, which is essential for performance [5, 3].

Six 195-cell libraries were designed under the ground rules of imec's A10 HD and HP nanosheet technology for physical design experiments with each PDN type (Table 1). While BS TSVM and FS M0 share the same cell height, BS BSC library exhibits a more compact cell design. Power rails in BSPDN libraries are on the first BS layer BSM0 whose dimensions are shown in Table 1.

PDN designs of FS and BS PDNs are shown in Fig. 2. A 4-layer backside metal stack (Fig. 2(b)) is designed for BSPDNs, of which BSM1 to BSM3 have the same dimensions as in [4]. The block-level benchmarking methodology is described in Fig. 3, using a vector execution unit of a 64-bit ARM™ high-performance computing core as a representative benchmark for high-performance demand. Cadence® tools were used for physical implementation [6-7], with frequency targets chosen to represent a wide range of optimization requirements, from low (F_1) to high (F_5). The aim of the physical design

experiments pivoting around the core area targeting a certain frequency is to increasingly constrain the floorplan until the design can no longer be implemented without a quality of results (QoR) degradation. The QoR criteria include achieved frequency (calculated from worst negative slack), number of DRC violations and number of violated paths.

Block-level PPA

Power: Using a commercial power integrity tool [7], dynamic power and IR drop simulations were performed. Fig. 4 and Fig. 5 show the dynamic IR drop heatmaps and their cumulative distribution of different PDNs in HD and HP at their maximum frequencies. In both HP and HD, BSPDNs improve dynamic IR drop compared to FS M0 PDN, effectively enhancing the power efficiency. The dense PDN with a stripes' pitch of 18CPP is required for HP to achieve a similar IR drop as its' HD counterpart. As such 7T FS M0_18CPP is taken as the baseline for HP in further performance and area analysis.

Performance: HP libraries achieve a maximum frequency (F_5) that is 22% higher than that of HD libraries (F_3), enabled by their larger sheet width and correspondingly stronger devices (fig. 6).

Area-Performance trade-off: Fig. 6 also captures area scaling trends of different PDN options. In both HP and HD, BSPDNs enable smaller core area than FS M0 PDN, while BS BSC achieves smallest core area overall. When frequency is pushed further towards high performance, an area gap develops between the FS M0 PDN reference and the BS TSVM, despite their identical cell area. BS TSVM allows more nets to be completed at lower metal layers compared to FS M0 PDN, benefiting area-efficient performance (Fig. 7). Furthermore, BS BSC obtains greater area reduction (from 26% to 39%) over FS M0 PDN when increasing the frequency in HP as efficient signal routing is maintained. Interestingly, block-level area reduction exceeds standard cell area scaling (BS TSVM to BS BSC), pointing to additional gains from a shorter interconnect (Fig. 8). Area reduction of BS BSC over FS M0 PDN in HD becomes smaller (from 23% to 13%) when increasing frequency (Fig. 6). Block-level area reduction from BS TSVM to BS BSC is less than cell scaling (Fig. 8). Both are accompanied by lower utilization for BS BSC (-19%, Fig. 9(a)). Indeed, scaling standard cells to 4.5T in HD has increased the pin density beyond a tipping point for a higher frequency target (e.g., F_3 , Fig. 9(b)). Consequently, routing resource demand has become so high that high utilization is difficult to maintain (Fig. 9(c)).

Conclusions

With physical design experiments and power integrity analyses, we have evaluated block-level PPA benefits of two BSPDN options TSVM and BSC over the traditional frontside power delivery in A10 HP and HD technology scenarios. Both BSPDN options are effective in boosting the core area scaling for high-performance designs in HP and HD. This is because moving the PDN to the wafer's backside frees space at the frontside which benefits signal routing. Scaled BSC libraries achieve the best PPA in both HP and HD, while it is more beneficial in HP than in HD with the densest 4.5T HD results showing signs of routing congestion at high frequencies.

References

- [1] J. Ryckaert et al., EDTM 2019. [2] D. Prasad, et al. IEDM 2019 [3] S. Yang et al., VLSI 2023. [4] G. Sisto et al., VLSI 2023 [5] S.C. Song et al., VLSI 2022 [6] Cadence® Innovus™ Implementation System [7] Cadence® Voltus™ IC Power Integrity Solution

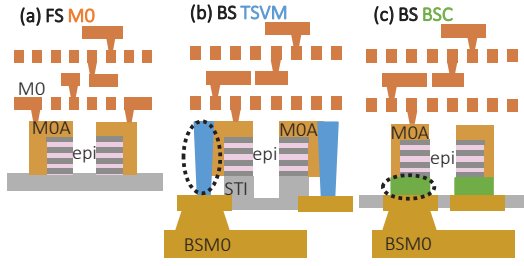


Fig. 1 Frontside (FS) and Backside (BS) power delivery (PD) options, where the power rails are on M0 layer (FS M0) and the first BS metal layer BSM0 (BS TSVM and BS BSC), respectively.

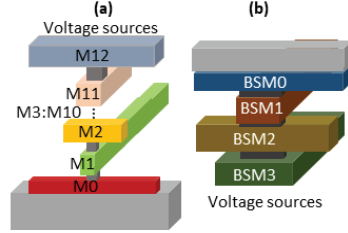


Fig. 2 PDN distribution over metal layers. (a) is FS M0 PDN, and (b) is BSPDN.

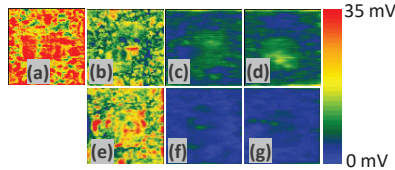


Fig. 4. Dynamic IR drop heatmaps for different PDNs: (a) 36CPP pitch 7T M0 FS PDN, (b) 18CPP pitch 7T M0 FS PDN, (c) 7T BS TSVM, (d) 5.5T BS BSC, (e) 36CPP pitch 6T FS M0 PDN, (f) 6T BS TSVM, (g) 4.5T BS BSC. All plots share the same scale (0-35mV) and have been obtained relying on a transient power simulation over a period of one nanosecond, with 5% switching activity for both inputs and flops.

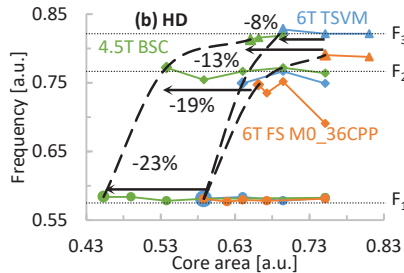
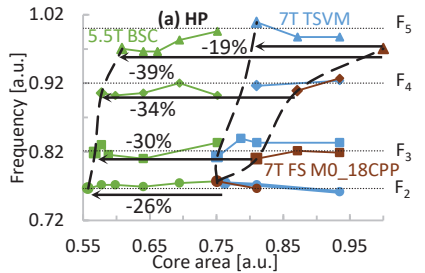


Fig. 6 Performance-area plots from physical implementation results of different PDN options at different frequencies in HP and HD. Dotted lines represent the frequency targets. Smoothed long-dash lines indicate minimum core area datapoints for BSC, TSVM, and FS M0 PDNs with increasing frequency.

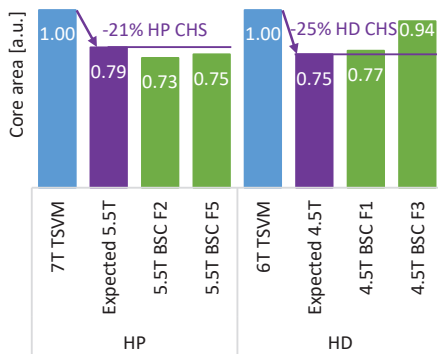


Fig. 8 Area scaling comparison between HP and HD scenarios at their two extreme frequencies with BSPDN TSVM as the baseline for each scenario. Expected cases are based on the cell height scaling (CHS) from TSVM standard cells to BSC ones, i.e., 5.5T/7T for HP and 4.5T/6T for HD. The area scaling potential of BS BSC is harnessed better in HP than in HD as the core area scaling for HP at the maximum frequency is better than that of HD.

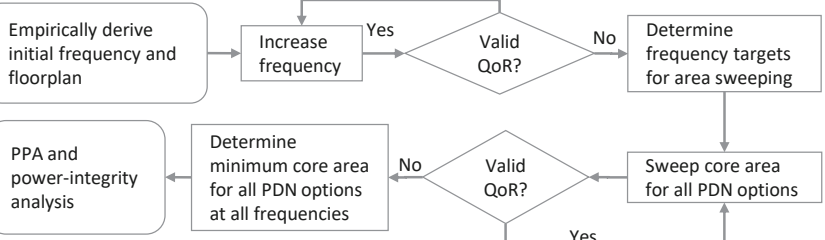


Fig. 3 Flowchart of our block-level PPA evaluation methodology of FS and BS PDN options

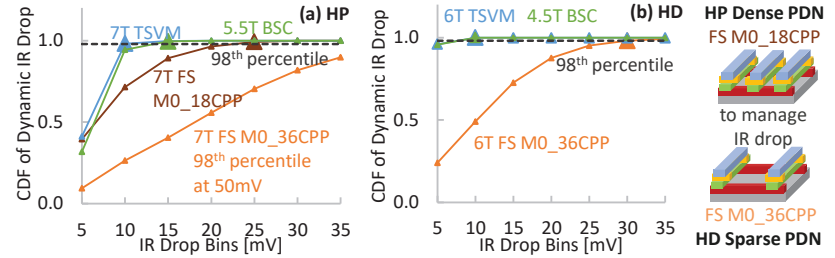


Fig. 5 Dynamic IR drop comparison of different PDN options in HP and HD. 18CPP PDN pitch improves IR drop of 7T FS M0 PDN. 7T FS M0_18CPP is thus used as the baseline for high-performance PPA comparison as it enables similar IR drop to that of 6T FS M0_36CPP.

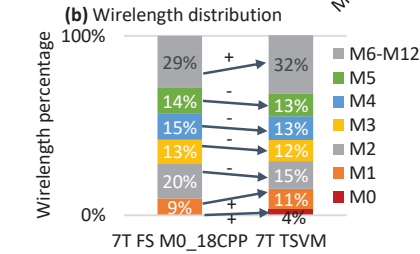
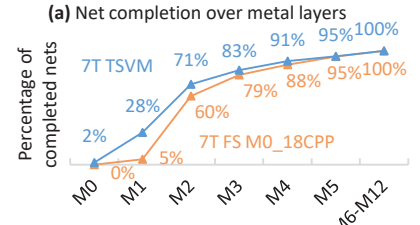


Fig. 7 (a) Net completion and (b) wavelength distribution comparison between 7T TSVM and 7T FS M0_18CPP at the latter's minimum core area targeting F_3 . Moving the PDN to the backside allows more nets to be completed at lower metal layers because it frees spaces for the 7T TSVM implementation to use more M0-M1 and M6-M12 metals and fewer M2-M5 metals for timing optimization.

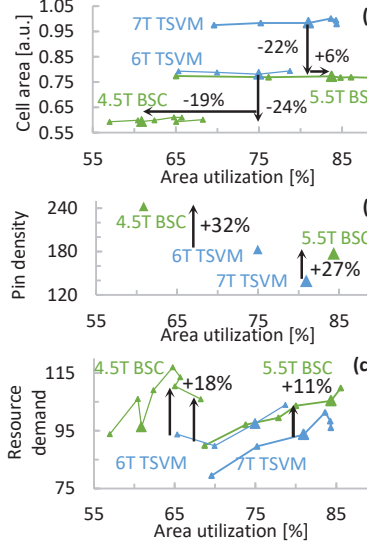


Fig. 9 (a) Cell area, (b) Standard cell pin density, and (c) Routing resource demand vs. area utilization of BSPDN TSVM and BSPDN BSC in HP (F_3) and HD (F_3). The cell area stably scales in line with CHS in both HP and HD at different utilization. However, core area scaling in HD from 6T TSVM to 4.5T BSC is worse than that of HP as the utilization of 4.5T BSC drops by 19%, while 5.5T BSC increases the utilization by 6% in HP. It is because the pin density of 4.5T BSC is increased beyond a tipping point for the high frequency target, resulting in the very high routing resource demand that causes routing difficulties.